



# A Supervised Learning Based Approach to Predict Solar Activity in the Temporal Domain

Snehasri Ravishankar<sup>1,3</sup>, Ravishankar B.T.<sup>2</sup>, Manju Sudhakar<sup>2</sup>, Bernard Foing<sup>3</sup>

<sup>1</sup>International Space University,

<sup>2</sup>Space Astronomy Group, URSC, Indian Space Research Organisation,

<sup>3</sup>LUNEX EuroMoonMars

## Introduction/Motivation

Many methods have been proposed towards the prediction of solar activity.

Primarily, whole 2D magnetograms have been used for this purpose. We show that using light-curves reduces the dimensionality of the problem, providing useful and interesting results. Using data from GOES and XSM, prediction of flares, background and event timing accurate to 99% has been demonstrated.

The data-driven nature of this model allows it to be applied on wavelengths over the EM spectrum.

We aim to extend the testing and applications to a wider range of data from other missions, and test ground and onboard intelligence for near-real time forecasting of activity and background.

## Data and Methodology

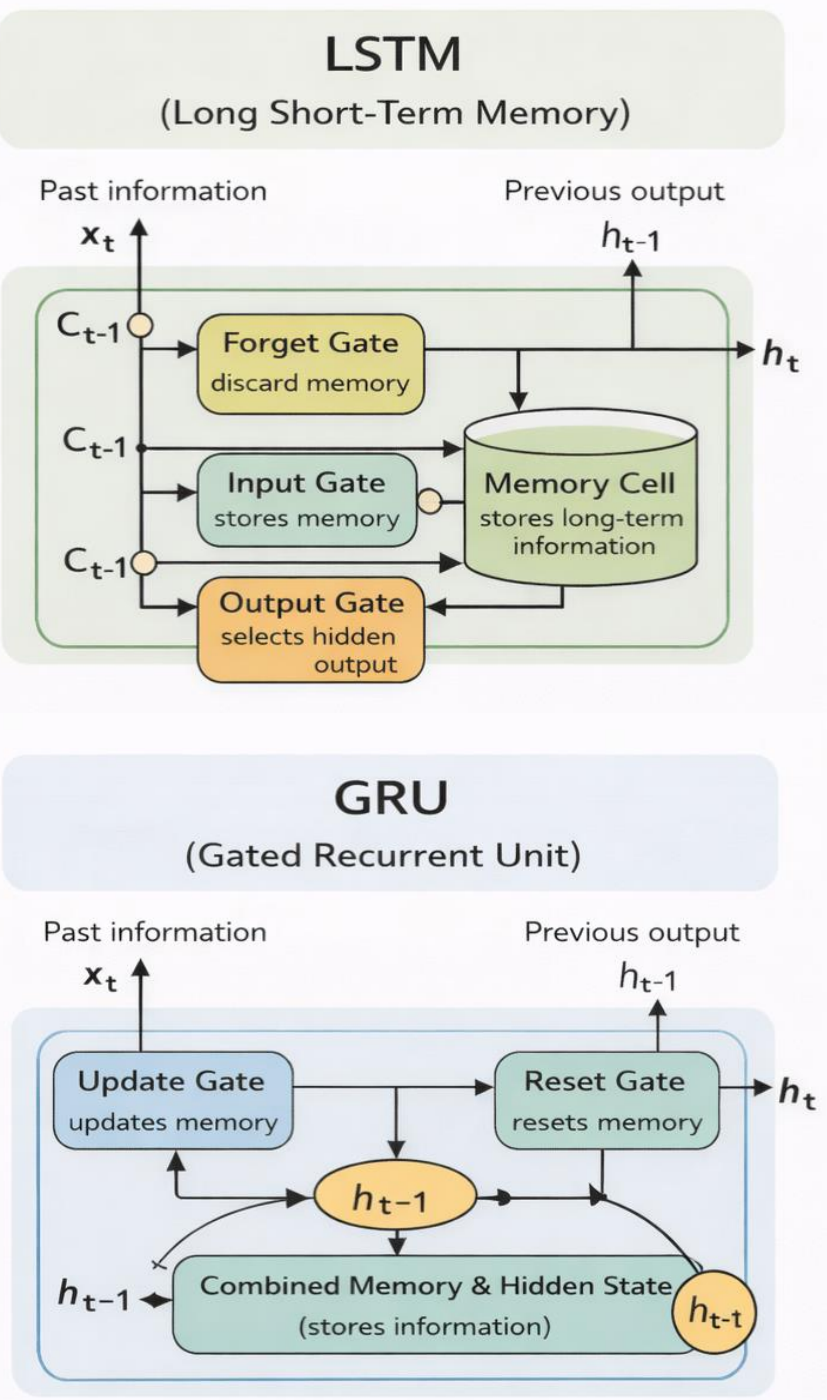
### • GOES-R XRS :

Science quality data of 1-min cadence

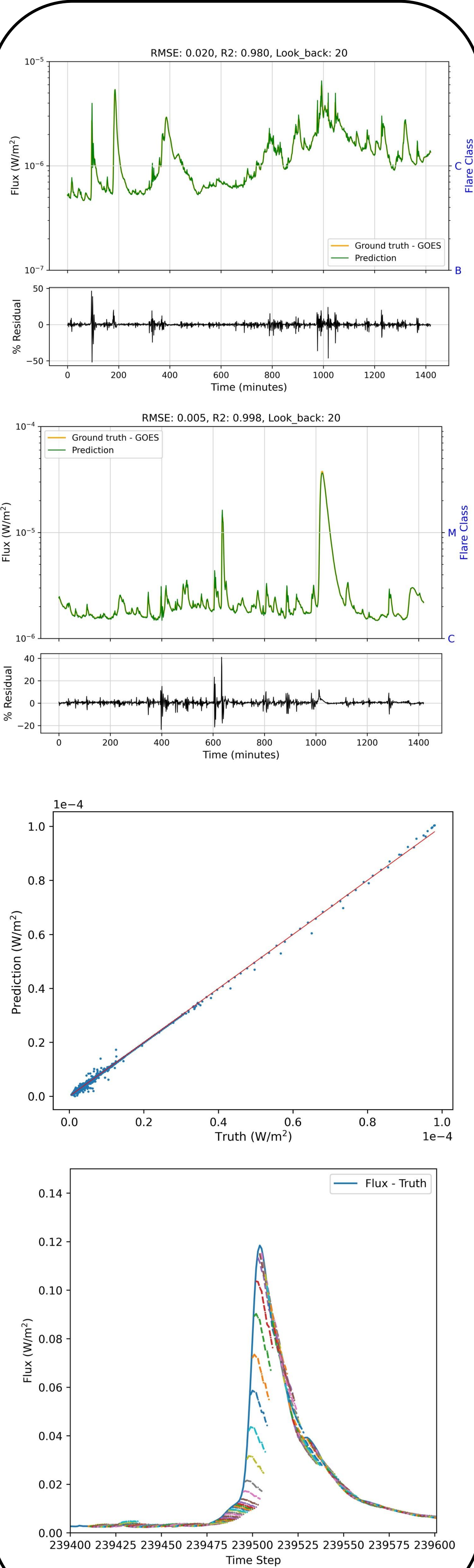
### • Also tested with Chandrayaan-2 XSM data:

1–15 keV soft X-ray spectrum, 1s cadence, energy resolution <180 eV @ 5.9 keV; raw/unprocessed with gaps requiring interpolation; 86,400 s (1 day) per file.

- LSTMs address the long-term dependency / vanishing-gradient limitation of standard RNNs via feedback connections
- Upon comparison of this main model with a lighter-weight and more resource-efficient network (GRU), we find the GRU provides comparable predictions and would thus be more amenable with CubeSat applications and onboard processing - where resource usage becomes critical.



## Model test – Metrics across Flare Class



## Near-Real Time Prediction

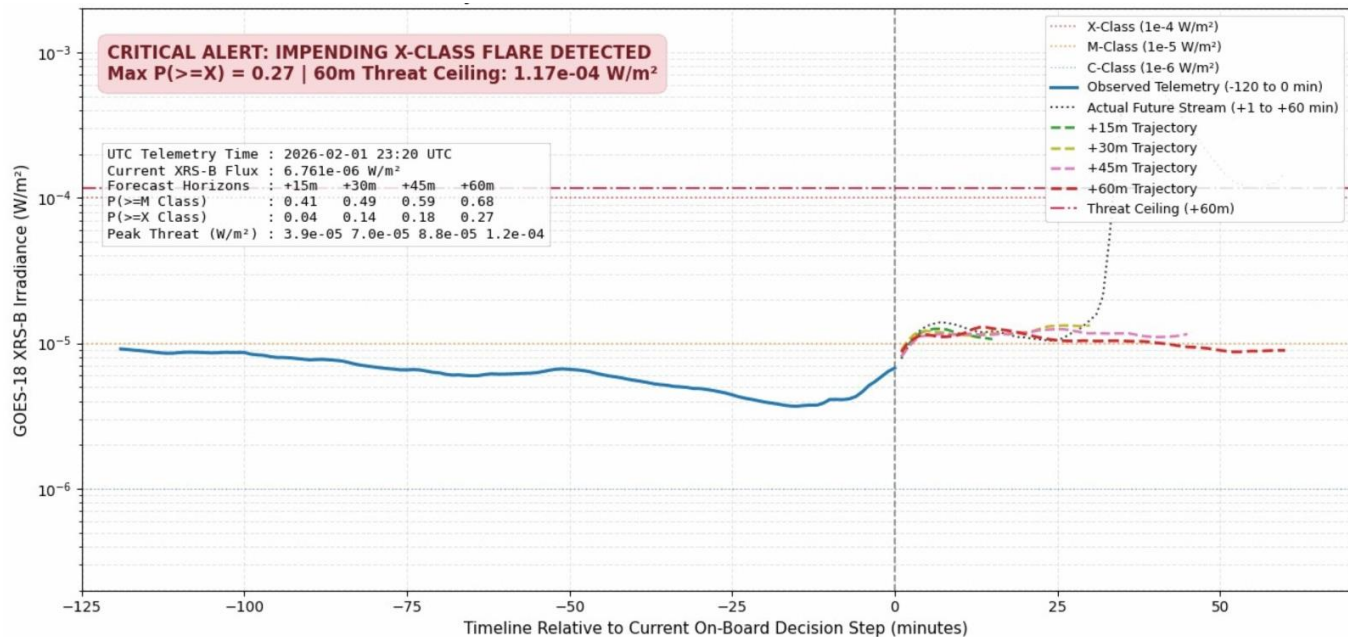
With multi-step forecasting, there is always a trade-off between future length and accuracy.

To overcome the above limitation, physics-based inputs using multi-instrument, multi-year training data – X-ray and EUV (from GOES) are included.

Temporal Convolution Network (TCN) with two independent channels to capture flat and impulsive spikes is introduced;

Trained using a 2-hour lookback and multiple horizon forecasts of 15, 30, 45 and 60 minutes with on-board warning for impending flares.

## In Progress and Future Work



### On-Board Telemetry Stream & Real-Time Multi-Horizon Forecast - 01/02/2026 23:20 UTC

| Horizon | $\log_{10}R^2$ | C-Class<br>TSS /<br>Recall | M-Class<br>TSS /<br>Recall | X-Class<br>TSS /<br>Recall | Threat<br>Coverage |
|---------|----------------|----------------------------|----------------------------|----------------------------|--------------------|
| +15 min | 0.9034         | 0.694 /<br>69.7%           | 0.715 /<br>94.3%           | 0.653 /<br>72.2%           | 88.08%             |
| +30 min | 0.7991         | 0.652 /<br>65.7%           | 0.647 /<br>91.8%           | 0.685 /<br>79.1%           | 84.48%             |
| +45 min | 0.7135         | 0.629 /<br>63.3%           | 0.587 /<br>91.1%           | 0.684 /<br>86.1%           | 86.28%             |
| +60 min | 0.6391         | 0.604 /<br>60.9%           | 0.461 /<br>92.8%           | 0.674 /<br>92.9%           | 88.03%             |

## Conclusions

- We can forecast even better with a better rate of data input and intelligence that could facilitate live prediction. Without a rolling window of updating detected measurements, it is required to forecast up to a point and save for comparison. The results indicate that longer-term live prediction could be a reality with some improvements with this approach of multi-step single-shot prediction.
- Short-term forecasting can be part of inbuilt intelligence of space-based systems – often resource limited - by efficiently changing modes of operation based on predicted solar or stellar source activity.
- Long-term predictions of activity spanning weeks to months can help mission planning through different observation cycles of space-and ground-based observatories.

Please get in touch! : sr576@snu.edu.in